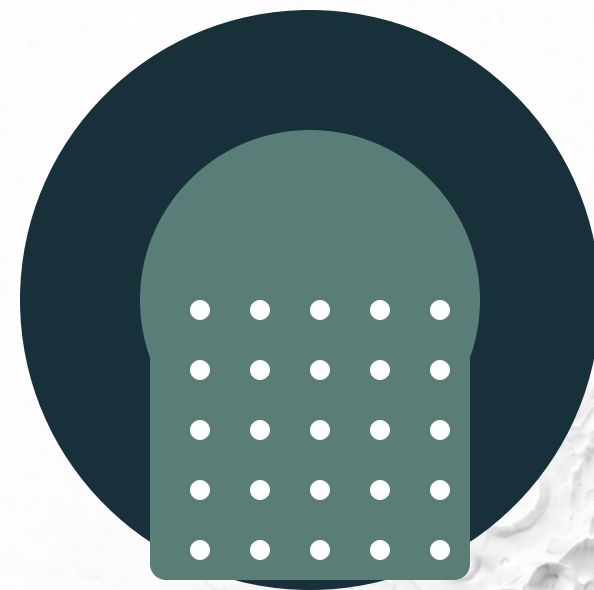


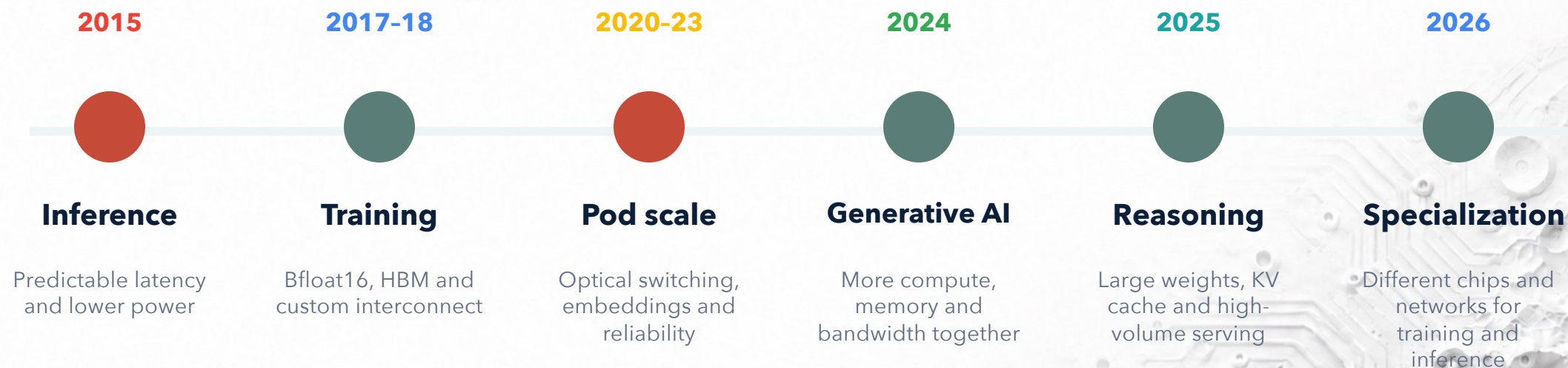
Google TPU

From inference ASIC to AI supercomputer

History, architecture, generation-by-generation design choices,
and the 2026 roadmap



The TPU evolved by attacking one bottleneck at a time



The recurring design question

Which resource prevents useful model work from scaling: arithmetic, memory, communication, cooling, software, or productive uptime?

Google built TPU v1 to avoid doubling data-center compute

2013 capacity projection

If deep-neural-network speech recognition reached broad use, Google estimated that serving demand could require roughly twice its data-center compute capacity.

2x

projected compute footprint



Matrix arithmetic

Neural networks spend most of their time multiplying and accumulating large tensors.

Tail latency

Voice search needs nearly every request to finish quickly, not only a good average.

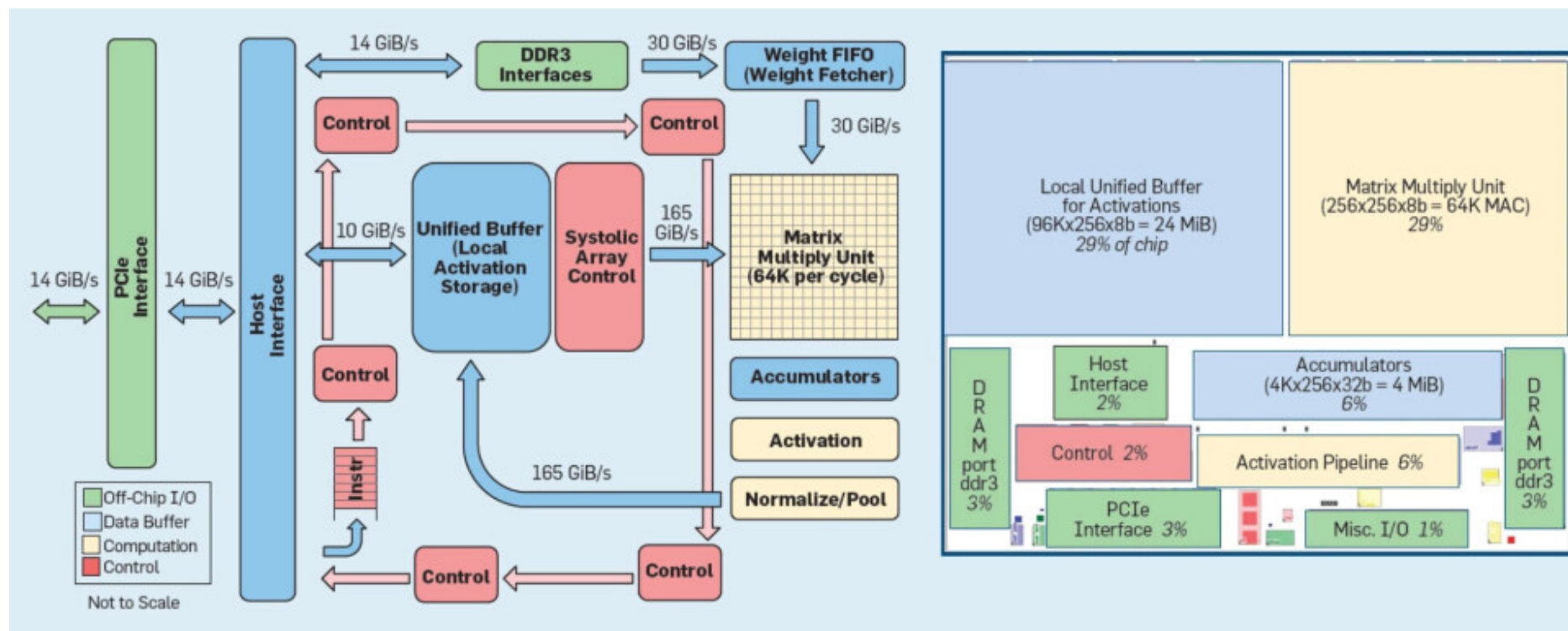
Power

More servers also require more electricity, cooling and floor space.

Design response: dedicate silicon to predictable tensor computation

Historical context: the original comparison used Intel Haswell CPUs and Nvidia K80 GPUs, not modern accelerators.

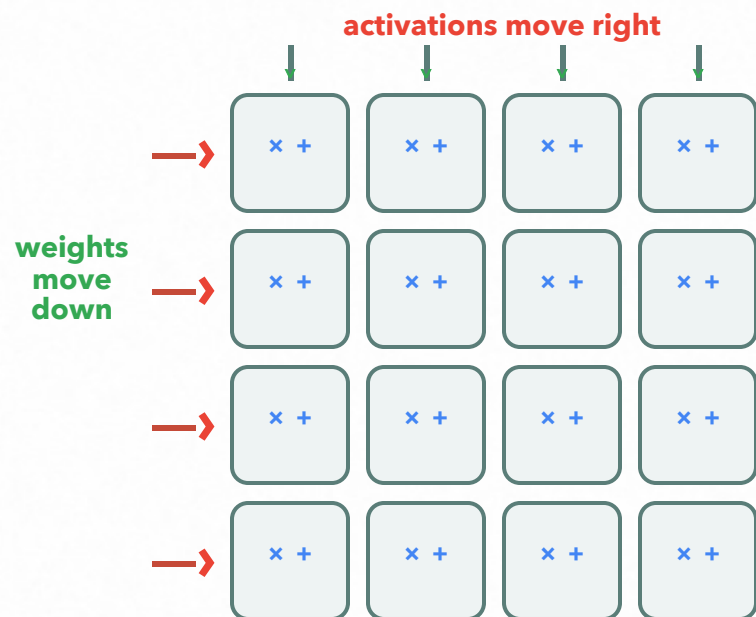
Google built TPU v1 to avoid doubling data-center compute



Historical context: the original comparison used Intel Haswell CPUs and Nvidia K80 GPUs, not modern accelerators.

A systolic array reuses data while it moves across the chip

A matrix-multiply unit (MXU) contains a grid of multiply-and-accumulate cells. Each value moves to a neighboring cell and participates in several calculations before leaving the array.



Why this saves energy

- A value fetched from memory is reused by many cells.
- Partial sums stay inside the array while the matrix product is formed.
- The chip spends less time and energy moving data through general-purpose caches.

Modern TPU TensorCores add vector and scalar units for operations such as softmax, normalization, addressing and control.

TPU v1 solved production inference

TPU v1

92 TOPS

peak 8-bit integer
operations

256×256

systolic multiply array

28 MiB

software-managed
on-chip memory

2015

deployed inside
Google data centers

Reduced precision

INT8 arithmetic used fewer transistors and less energy per operation.

Deterministic execution

A simpler machine better matched strict 99th-percentile latency targets.

Measured result

Google reported 15-30× higher performance and 30-80× higher performance per watt than its contemporary CPU and GPU systems.

Exposed bottleneck

Slower external memory prevented some workloads from using the full arithmetic rate.

The comparison is historically important, but it does not compare TPU v1 with a current GPU.

TPU v2 and v3 turned the accelerator into a training supercomputer

TPU v2 | 2017

Training required floating-point range, fast model memory and direct communication between chips.

45 TFLOPS

BF16 per chip

16 GiB

HBM per chip

256 chips

historical pod

2D torus

custom ICI network

TPU v3 | 2018

Higher compute and twice the HBM capacity enabled larger models, but the denser system required liquid cooling.

123 TFLOPS

BF16 per chip

32 GiB

HBM per chip

1,024 chips

per pod

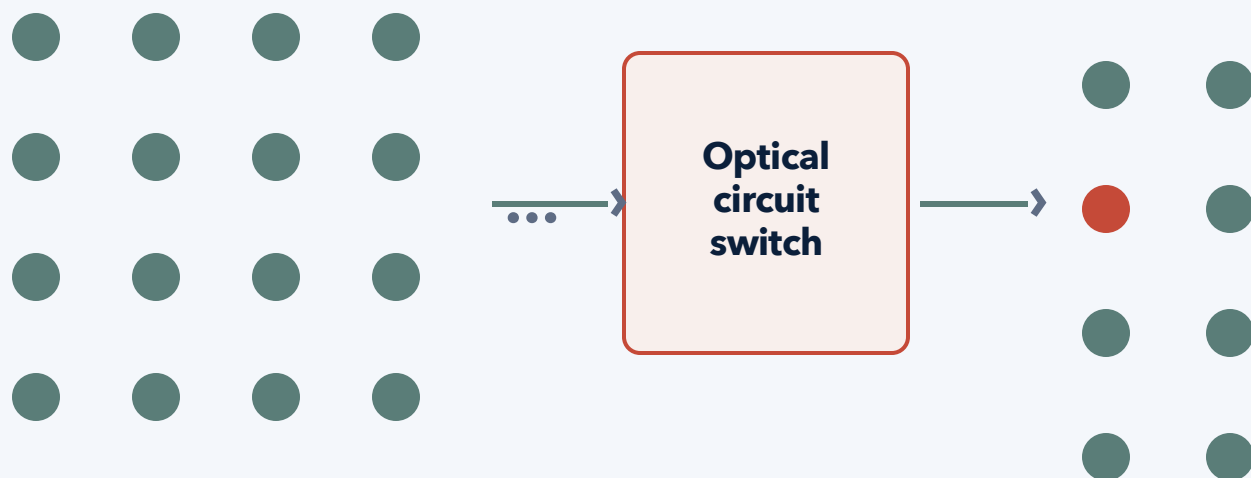
Liquid

cooling for sustained density

Bfloat16 keeps the exponent range of FP32 but uses fewer fraction bits, reducing arithmetic and memory cost while accumulating results in FP32.

TPU v4 made networking and embeddings first-class problems

64-chip cube



A failed link or cube can be bypassed by reconfiguring the optical fabric.

275

BF16 TFLOPS per chip

4,096

chips per pod

1.1

peak BF16 exaflops per pod

SparseCore accelerated embedding-heavy models by 5-7× in Google's evaluation while using about 5% of die area and power.

TPU v5 split cost efficiency from maximum performance

TPU v5e

Efficiency and flexibility

197

BF16 TFLOPS per chip

16 GB

HBM per chip

256

chips per pod

1-256

supported slice sizes

Best fit: medium- and large-scale training or serving where performance per dollar matters more than maximum single-pod scale.

TPU v5p

Frontier training performance

459

BF16 TFLOPS per chip

95 GiB

HBM per chip

8,960

chips per pod

1.2 TB/s

bidirectional ICI per chip

Best fit: large foundation models and embedding-heavy workloads that need maximum memory, interconnect bandwidth and pod scale.

Trillium increased compute, memory and communication together

4.7
×

Peak compute

918 BF16 TFLOPS per chip through larger 256×256 MXUs and a higher clock rate.

2×

HBM capacity and bandwidth

32 GB at 1.638 TB/s supports larger weights and larger attention key-value caches.

2×

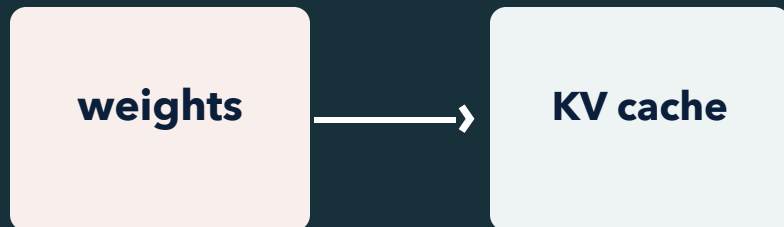
Inter-chip bandwidth

800 GB/s bidirectional ICI improves scaling inside the 256-chip pod.

A 256-chip pod is the fast local domain. Third-generation SparseCore handles fine-grained embedding access, while Multislice connects many pods when a job exceeds that boundary.

Ironwood attacks the memory wall in training and inference

When a model generates one token at a time, the chip repeatedly reads weights and the attention key-value cache. Computation can wait for memory even when the MXUs are extremely fast.



Ironwood keeps far more of both near the compute units.

192 GiB

HBM per chip

7.38 TB/s

HBM bandwidth

4,614

FP8 TFLOPS per chip

9,216

chips per superpod

42.5

peak FP8 exaflops per pod

2×

performance per watt vs
Trillium

Google Cloud documentation positions TPU7x for dense and mixture-of-experts models, pre-training, sampling and decode-heavy inference.

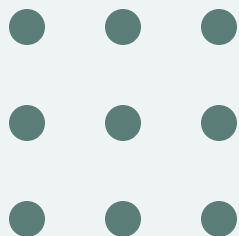
TPU 8 uses different architectures for training and inference

Announced in April 2026; Google still describes both systems as coming soon

TPU 8t

Large-scale pre-training

3D torus



12.6 FP4 PFLOPS
216 GB HBM
SparseCore +
decoder engine
9,600-chip superpod

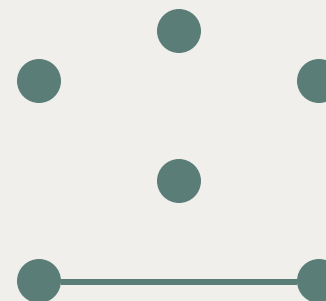
Regular neighbor-heavy
communication

FP4 and direct storage paths reduce data movement; reliability
features target more than 97% useful training time.

TPU 8i

Serving and reasoning

Boardfly

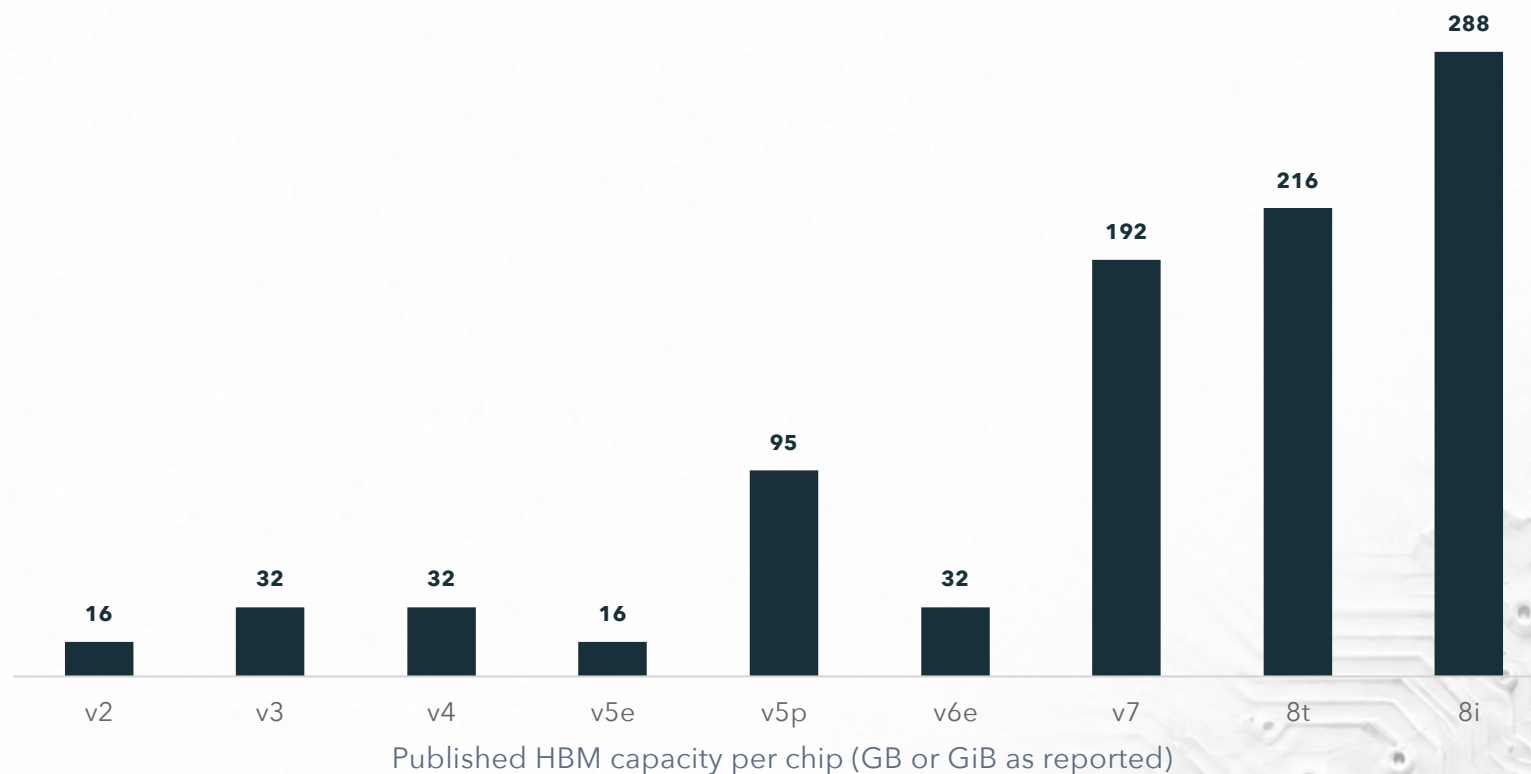


10.1 FP4 PFLOPS
288 GB HBM
384 MB on-chip SRAM
Collectives
Acceleration Engine

Flatter paths for all-to-all token
routing

Boardfly cuts the maximum path in Google's 1,024-chip example
from 16 hops to seven, reducing communication latency.

HBM capacity reflects workload specialization



How to read the pattern

v5e and v6e prioritize cost and versatile 256-chip pods, so their memory capacity is lower than the performance-oriented v5p and v7 products.

TPU 8i has the largest HBM and three times Ironwood's on-chip SRAM because low-latency reasoning must retain model state and larger KV caches close to compute.

Capacity alone does not predict speed. Bandwidth, precision, topology, compiler quality and utilization also matter.

TPU performance depends on the entire software and data path

Model code

JAX, PyTorch, TensorFlow or Keras

Compiler

XLA fuses operations, chooses layouts and generates TPU programs

Parallelism

Sharding maps model, data, sequence and expert dimensions onto chips

TPU pod

MXUs, vector units, HBM, SparseCore or CAE, and the ICI network

System

Host CPUs, storage, data-center network, scheduler, cooling and recovery

Profiling feeds back into model structure, tensor shapes, sharding and input design.

TPUs favor large, regular tensor workloads

Strong fit

- Large, regular matrix multiplications
- Stable tensor shapes and sufficient batch size
- Models that can be sharded across the available topology
- Input and storage systems that keep the accelerator supplied

Common performance losses

- Dynamic shapes that trigger recompilation
- Small batches that leave the MXU underused
- Irregular control flow or unsupported custom operations
- Input starvation, memory traffic or inter-chip communication
- Porting cost and dependence on Google Cloud's TPU ecosystem

Peak FLOPS describe a ceiling. End-to-end performance depends on how much of the system remains productive during the real workload.

Primary sources and suggested reading

2017	TPU v1 architecture and production evaluation	research.google/pubs/in-datacenter-performance-analysis-of-a-tensor-processing-unit/
2020	TPU v2 and v3 training-supercomputer architecture	gwern.net/doc/ai/scaling/hardware/2020-jouppi.pdf
2023	TPU v4 optical network and SparseCore paper	arxiv.org/abs/2304.01433
2023	TPU v5e and v5p documentation and announcements	docs.cloud.google.com/tpu/docs/v5e docs.cloud.google.com/tpu/docs/v5p
2024	Trillium announcement and v6e specifications	cloud.google.com/blog/products/compute/introducing-trillium-6th-gen-tpus
2025	Ironwood architecture and system-level co-design	docs.cloud.google.com/tpu/docs/tpu7x cloud.google.com/blog/products/compute/inside-the-ironwood-tpu-codesigned-ai-stack
2026	TPU 8t and TPU 8i architecture deep dive	cloud.google.com/blog/products/compute/tpu-8t-and-tpu-8i-technical-deep-dive

Product specifications and availability can change. Check the current Google Cloud TPU documentation before making procurement or deployment decisions.